Arc Lengths & Sector Areas

Exam code: 0607

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Written by: Naomi C
Reviewed by: Dan Finlay

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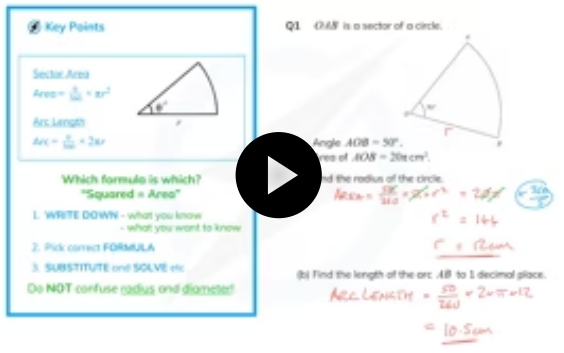

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Arc Lengths & Sector Areas

What is an arc?

- An **arc** is a **part** of the **circumference** of a circle
- Two points on a circumference of a circle will create **two arcs**
  - The **smaller** arc is known as the **minor arc**
  - The **bigger** arc is known as the **major arc**

What is a sector?

- A **sector** is the part of a circle enclosed by two **radii** (radiuses) and an arc
  - A sector looks like a slice of a circular pizza
  - The curved edge of a sector is the arc
- Two radii in a circle will create **two sectors**
  - The **smaller** sector is known as the **minor sector**
  - The **bigger** sector is known as the **major sector**

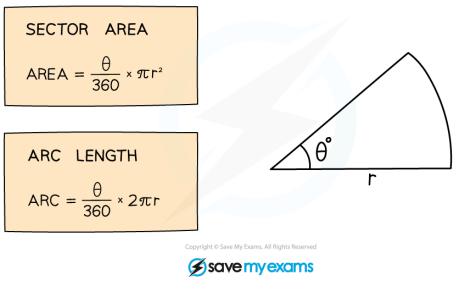
What formulae do I need to know?

- You need to be able to calculate the **length of an arc** and the **area of a sector**
- The **angle** formed in a sector by the two radii is often labelled θ (the Greek letter "theta")
- You can calculate the **area of a sector** or the **length of an arc** by adapting the formulae for the area or

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Area of a sector $= \frac{\theta}{360} \times \pi r^2$
- Arc length $= \frac{\theta}{360} \times 2\pi r$



- Working with s**ector and arc formulae** is just like working with any other formula:
  - **Write down** what you know (or what you want to know)
  - Pick the correct **formula**
  - **Substitute** the values in and **solve**

How do I find the length of an arc?

- **STEP 1**
  Divide the **angle** by **360** to form a fraction
  - $\frac{\theta}{360}$
- **STEP 2**
  Calculate the **circumference** of the **full circle**
  - $2\pi r$
- **STEP 3**
  Multiply the **fraction** by the **circumference**
  - $\frac{\theta}{360} \times 2\pi r$

How do I find the area of a sector?

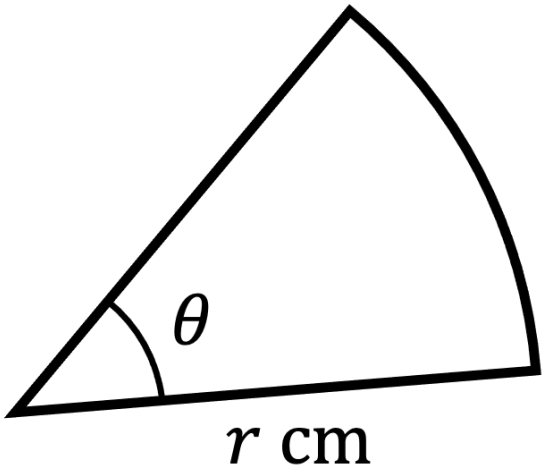
- **STEP 1**
  Divide the **angle** by **360** to form a fraction
  - $\frac{\theta}{360}$
- **STEP 2**
  Calculate the **area** of the **full circle**
  - πr^2
- **STEP 3**
  Multiply the **fraction** by the **area**
  - $\frac{\theta}{360} \times \pi r^2$

Examiner Tips and Tricks

fraction of the whole circle.

Worked Example

A sector of a circle is shown.



Answer:

The angle, θ, is 72° and the radius, r, is 5 cm.

(a) Find the area of the sector, giving your answer correct to 3 significant figures.

Substitute $\theta = 72°$ and $r = 5$ into the formula for the area of a sector, $A = \frac{\theta}{360}\pi r^2$

$$A = \frac{72}{360}\pi \times 5^2$$

Use a calculator to work out this value

15.70796...

Round your answer to 3 significant figures

15.7 cm²

(b) Find the length of the arc of the sector, giving your answer as a multiple of π.

Answer:

Substitute $\theta = 72°$ and $r = 5$ into the formula for the length of an arc, $l = \frac{\theta}{360}2\pi r$

$$l = \frac{72}{360} \times 2 \times \pi \times 5$$

Simplify the number part without π

Write down the final answer with π

2π cm

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